

MFT-04 (Multiple Choice Questions)

1. Turing machine was invented by: [BL-1]
 - a) Alan Turing
 - b) Turing man
 - c) Turing theory
 - d) Noam Chomsky
2. Turing machine is more powerful than: [BL-1]
 - a) Finite automata
 - b) Push down automata
 - c) Both (a) and (b)
 - d) None of these
3. In one move the Turing machine: [BL-2]
 - a) May change its state
 - b) Write a symbol on the cell being scanned
 - c) Move the head one position left or right
 - d) All of the above
4. A Universal Turing Machine (UTM) can simulate any other Turing Machine (True/False) [BL-1]
5. The Turing Church Thesis states that Turing Machines can solve any computable problem (True/False) [BL-1]
6. Find true or false statement. [BL-2]
 - a) Turing machine was developed by Alan Turing
 - b) PDA is less powerful than Turing machine
 - c) Both (a) and (b)
 - d) None of these

7. The value of n if Turing machine is defined using n -tuples: [BL-1]
- a) 3
 - b) 5
 - c) 6
 - d) $2L+1$
8. If δ is not defined on the current state and the current tape symbol, then the machine ____
- a) Does not halts
 - b) Halts
 - c) goes into loop forever
 - d) none of the mentioned
9. The Turing Church Thesis states that Turing Machines can solve any computable problem. (True/False) [BL-1]
10. The Language Accepted by Turing machine is called ____.
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Long Answer Questions

11. Construct a Turing Machine that can accept the string over $\{0,1\}$ containing even number of 1's. [BL-3]
12. Construct a Turing Machine to recognize the language: $\{ a^n b^n c^m \mid n, m \geq 1 \}$ [BL-3]
13. Design a TM to accept the language $L = \{ a^n b^n c \mid n > 1 \}$ [BL-3]
14. Design a TM to accept the set of all strings $\{0,1\}$ with 010 as substring. [BL-3]
15. Design a Turing Machine to accept the language $L = \{ 0^n 1^n \mid n \geq 1 \}$. [BL-3]

16. Explore the distinction between recursive and recursively enumerable languages. Define each type of language and discuss their properties, including decidability and recognizability. [BL-2]
17. Construct a Turing machine for $L = \{a^i b^j c^k \mid i * j = k, i, j, k \geq 1\}$. [BL-3]
18. Construct Turing machine for $L = \{a^n b^m a^{(n+m)} \mid n, m \geq 1\}$ [BL-3]
19. Explain the concept of a Turing Machine (TM) and how it serves as a fundamental model of computation. Describe its components, including the tape, read/write head, states, and transition function. [BL-2]
20. Construct TM for the addition function for the unary number system. [BL-3]
21. Implement a Turing machine that, given an input number **n** in binary representation, multiplies the number by 2. [BL-3]
22. Construct a TM for the language $L = \{(0^n 1^{2n}) \text{ where } n \geq 1\}$. [BL-3]
23. Construct a TM machine for checking the palindrome of the string of even length. [BL-3]
24. Construct a TM machine for checking the palindrome of the string of odd length. [BL-3]